ASSIGNMENTS

1. Define Artificial Intelligence. Write some applications of artificial intelligence.
2. Write down the goals of artificial intelligence.
3. Define Machine Learning. Describe three forms of machine learning (Supervised, Unsupervised and Reinforcement Leaning) with example.
4. Differentiate between Supervised Leaning and Unsupervised Leaning with example.
   or
   Differentiate between Classification and Clustering with example.

5. Consider the dataset below to make a decision tree to be learned which predicts whether people will pass AI (True or False), based on their previous GPA (High, Medium, or Low) and whether they studied.

GPA	Studied	Passed
L	F	F
L	T	T
M	F	F
M	T	T
H	F	T
H	T	T

Table 1

i. What is the entropy **H(Passed)**?

ii. What is the entropy **H(Passed | GPA)**?

iii. What is the entropy **H(Passed | Studied)**?

iv. Draw the decision tree with the root and its immediate successors that would be learned for this dataset. Put your logic for selecting the attribute as the root.

5. We have eight data points: X={2, 3, 4, 6, 10, 11, 12, 15}. Assume k = 2 and the initial centroids are $C_1 = 5$ and $C_2 = 8$. Perform one full iteration of K-means clustering. Write down the new clusters and their centroids.

6. What is Searching? Differentiate between uninformed search and heuristic search.
7. Differentiate between BFS and DFS.

8. Apply Best First Search and A* search on the following graph to find a path from source S to goal G.

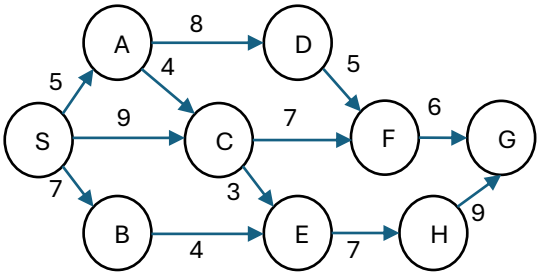


Fig. 1

H(n)

S	45
A	40
B	32
C	27
D	25
E	19
F	17
G	0
H	15

Table 2

9. Define Production System and Production Rules. Perform water-jug problem where We are given 2 jugs, a **4 liter** one and a **3 liter** one. Neither has any measuring markers on it. There is a pump that can be used to fill the jugs with water. How can we get exactly 2 liters of water into the 4-liter jugs?
10. Differentiate between Forward-chaining and Backward-chaining.
11. Differentiate between **Propositional Logic (PL)** and **First-Order Logic (FOL)**.
12. Convert the statements to PL or FOL expressions as needed. (From the corresponding slide)
13. Find whether the meaning of the statement is true or false: "If humans colonize Mars or aliens visit Earth, then Elon Musk might be an entrepreneur but wasn't an engineer."
14. Convert the following English statements into Propositional Logic (PL) or First-Order Logic (FOL) expressions as appropriate:
    i. If it is raining, then the street is wet.
    ii. Every student likes at least one teacher.
    iii. If it is daytime and it is not raining then it is warm.
    iv. Only humans can drive cars.
    v. Everyone has a friend.
15. Using Resolution Method, prove from the given knowledge base that Knows(Mike, Eleven) is true:
    Friend(x, y) → Knows(x, y)
    Friend(x, y) ∧ Knows(y, z) → Knows(x, z)
    Friend(Mike, Will)
    Friend(Will, Eleven)

16. Define Reasoning. Write down the types of Reasoning.
17. Explain conditional probability.
18. In a class, there are 70% of the students who like English and 40% of the students who likes English and mathematics, and then what is the percent of students those who like English also like mathematics?

19. In a school, 70% of the students play football and 30% do not. Among the football players, 40% are good at math, while among the non-football players, 20% are good at math. If a student is chosen at random, what is the probability that the student is good in math.

20. A dataset contains 2D points representing students' study hours (x) and their previous test scores (y). The class label is whether the student **passed (P)** or **failed (F)** in the final exam.

The training data is as follows:

Student	Study Hours (x)	Score (y)	Class
S1	2	50	F
S2	4	65	P
S3	5	80	P
S4	1	45	F
S5	6	70	P

A new student **S** has values:

- Study Hours = 3
- Score = 60

(a) Using **Euclidean distance**, calculate the distances from S? to all training points.

(b) For **K = 3**, determine the predicted class for S?. Show your steps clearly.

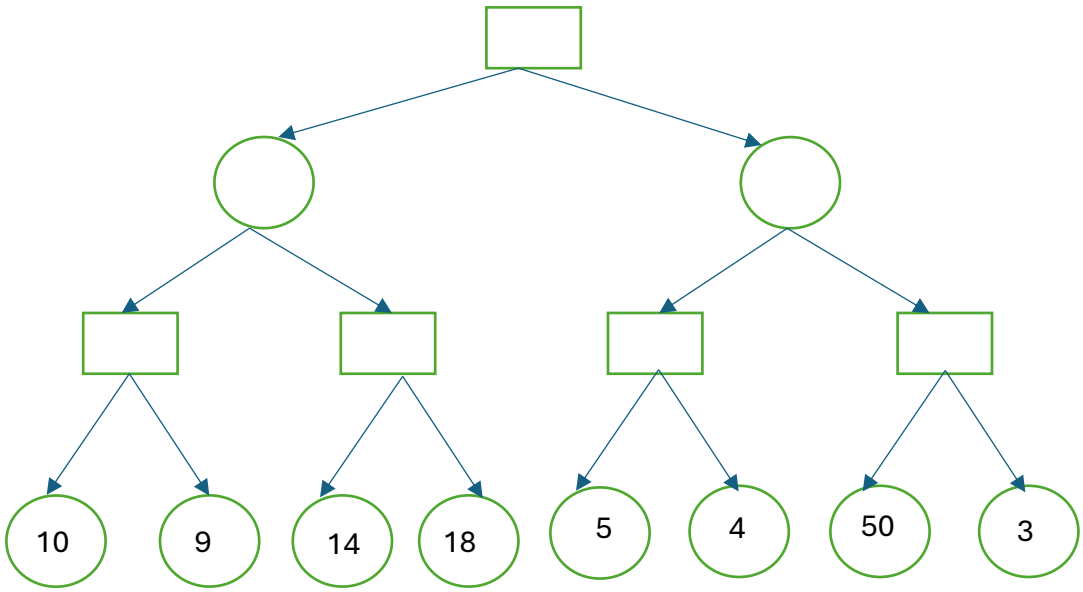
(c) Briefly explain one advantage and one disadvantage of K-NN.

21. In a university, 70% of the students are from the CSE department and 30% are from the EEE department. It is known that 40% of the CSE students play football, while only 10% of the EEE students play football. If a randomly selected student is found to be a football player, use Bayes Theorem to calculate the probability that the student belongs to the CSE department.

22. A factory has three machines—M1, M2, and M3—that produce 20%, 50%, and 30% of the total items respectively. The defective rates of the machines are 1% for M1, 3% for M2, and 2% for M3. If an item is chosen at random and it is found to be defective, calculate the probability that the item was produced by machine M2.

23. On a new day, the weather is sunny, and the temperature is mild. Using the Naïve Bayes classifier, calculate the probability that people will decide to play, and the probability that they will not play. Finally, determine which decision will be predicted.

Weather	**Temperature**	**Play**
Sunny	Hot	No
Sunny	Mild	No
Overcast	Hot	Yes
Rainy	Mild	Yes
Rainy	Cool	Yes
Sunny	Mild	Yes

24. In an email classification system, 40% of all emails are spam. The word **"Offer"** appears in 70% of spam emails but only in 10% of non-spam emails. Similarly, the word **"Free"** appears in 60% of spam emails but only in 5% of non-spam emails. If a new email contains both the words "offer" and "free," use Naïve Bayes to calculate the probability that the email is spam and the probability that it is not spam. State which category the classifier will choose.

25. Consider the following two player game tree in which the static scores are given from the first player's point of view. Apply the **Min-Max Search Algorithm** and compute the value of the root of the tree. Also find the most convenient path for Max node.